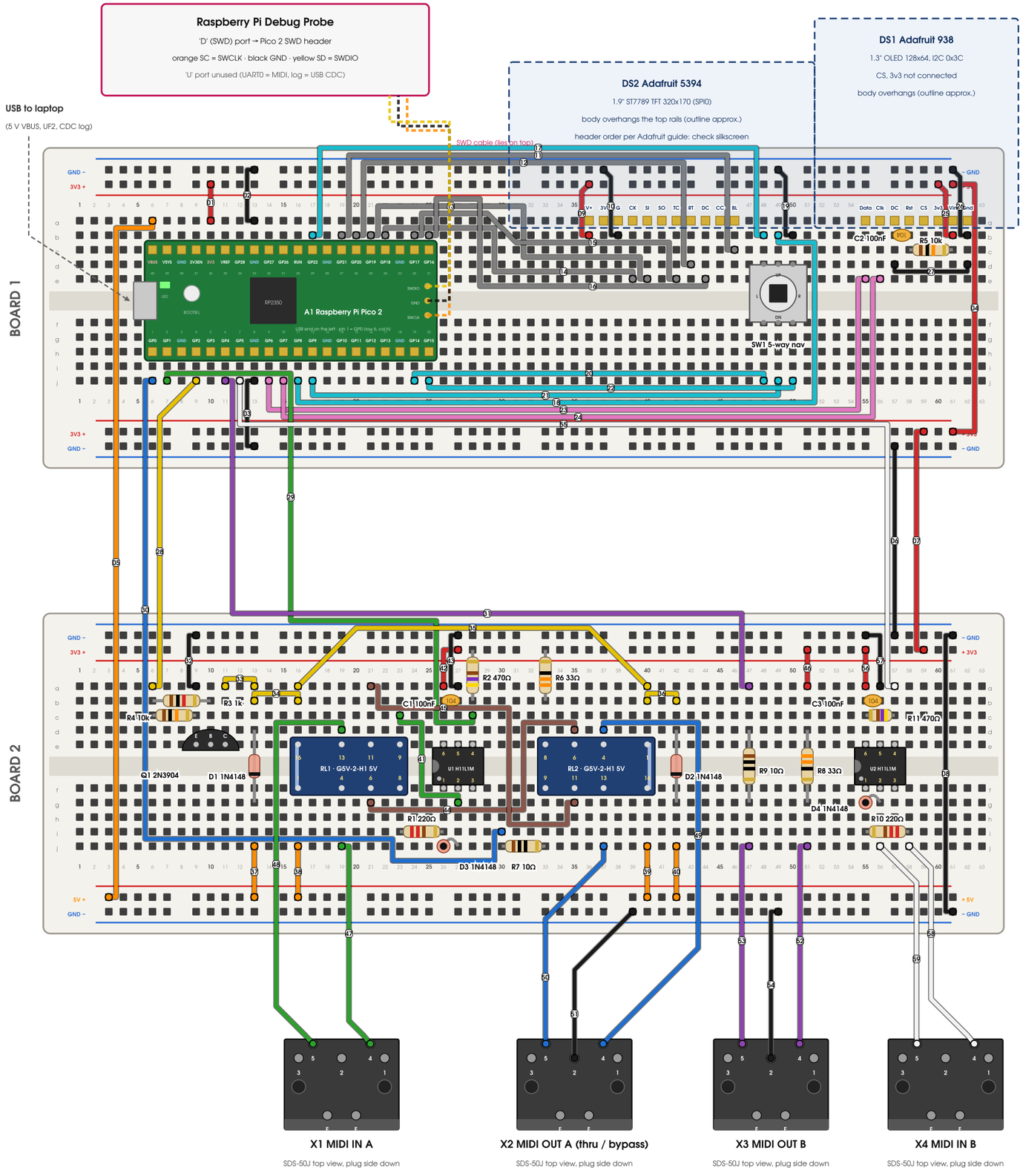
Sheet 1 · Breadboard layout, top view, drawn on a 0.1" grid (scale 1.26:1)



Wire colours

- 3V3
- 5V (VBUS)
- GND
- MIDI IN A path (DIN IN A, U1, GP1)
- MIDI IN B path (DIN IN B, U2, GP5)
- MIDI OUT A (GP0, DIN OUT A)
- MIDI OUT B (GP4, DIN OUT B)
- Bypass links (relay NC to NC)
- Relay drive (GP2, coil -)
- I2C1 to OLED (GP6 SDA, GP7 SCL)
- SPI0 / control to TFT
- Joystick (GP8/9/14/15/22)

Symbols

Module body on its header (wires may pass under it)

Wire end: plug into this hole. 12 = wire W12 in the netlist

Rails: outer = GND (blue), inner = supply (red)
B1 inner rails 3V3 · B2 top inner 3V3, bottom inner 5V

Read me first

Rev B: adds MIDI IN B (X4, U2 rows 55-57, GP5), wires W55-W59.
W01-W53 as Rev A; W54 moves to the GND rail @49 (X3 moved).
Board 1 = MCU + UI. Board 2 = MIDI IN A/B, relays, OUT A/B.
Pico 2: USB end off the left edge of B1, pin 1 (GP0) at h6,
pin 40 (VBUS) at c6. Free holes per Pico pin: a/b and i/j.
B2 bottom inner rail is 5 V (VBUS) - relay coils only.
Everything else runs from 3V3 (Pico pin 36).
Check rail continuity at the middle of each rail with a meter:
some full-size boards split the rails between rows 31 and 33.
Resistors lie flat over the holes they span (unused holes).
D3 stands upright j26 (cathode, band) / j27; D4 the same, g55 / g56.
IN B has no bypass relay: it goes straight to R10 / U2.
Relay pin numbers are the Omron G5V-2 terminal numbers;
RL2 is rotated 180° (coil end on the right).
DIN sockets: short flying leads soldered to the socket pins.
IN A/B shell (E) and pins 1, 2, 3 stay unconnected.
OLED DC is tied to GND to force I2C address 0x3C (see sheet 3).
Both displays can stay fitted: firmware uses the OLED if it answers
at 0x3C, else the TFT. To test the TFT, pull W25 (OLED Vin).
Joystick: as drawn, push towards board-1 row a = UP.
Directions follow the ALPS SKQUCA pinout - verify with a meter.
Every connection was machine-checked against docs/wiring.md.

Off-board flying leads (DIN sockets)

Wire	Socket pin	Plugs into	Function
47	X1 MIDI IN A pin 4	B2 j19	RL1 COM1 (pin 4)
48	X1 MIDI IN A pin 5	B2 d19	RL1 COM2 (pin 13)
49	X2 MIDI OUT A pin 4	B2 d37	RL2 COM1 (pin 4)
50	X2 MIDI OUT A pin 5	B2 j37	RL2 COM2 (pin 13)
51	X2 MIDI OUT A pin 2	B2 bottom GND rail @39	GND
52	X3 MIDI OUT B pin 4	B2 j51	R8 33R
53	X3 MIDI OUT B pin 5	B2 j47	R9 10R
54	X3 MIDI OUT B pin 2	B2 bottom GND rail @49	GND (Rev B: was @53)
58	X4 MIDI IN B pin 4	B2 j58	R10 220R
59	X4 MIDI IN B pin 5	B2 j56	H11L1 U2 pin 2

X1/X4 MIDI IN A/B: pins 1, 2, 3 and the shell tabs (E) stay unconnected. X2/X3: pins 1, 3 unconnected.
Pin 2 is the shield pin. Front, top and solder-side views of the socket are on sheet 3.

MIDIum MT-1	melling.audio
Breadboard prototype · Layout (top view, milestone 1)	Sheet 1 of 4
Rev C · 19/10/2026 · J Melling	A3 · milestone 1

Sheet 2 · Jumper list: tick each row as you fit it

Hole names: B1/B2 = board, letter + number = column + row (as printed on the board). Rails are named by what they carry; @n = the rail hole next to row n. Length = routed length on sheet 1 (cut ~10 mm longer for bends).

	W	From	To	mm	Colour	Signal / purpose
	Power rails, board 1					
☐	**W01**	B1 a10	B1 top 3V3 rail @10	6	red	3V3 from Pico pin 36 to B1 top 3V3 rail
☐	**W02**	B1 a13	B1 top GND rail @13	11	black	GND from Pico pin 33 to B1 top GND rail
☐	**W03**	B1 j13	B1 bottom GND rail @13	14	black	GND from Pico pin 8 to B1 bottom GND rail
☐	**W04**	B1 top 3V3 rail @61	B1 bottom 3V3 rail @61	51	red	3V3 rail link, B1 top to B1 bottom
	Power between the boards					
☐	**W05**	B1 a6	B2 bottom 5V rail @3	126	orange	VBUS (Pico pin 40) to B2 bottom 5V rail
☐	**W06**	B1 bottom GND rail @57	B2 top GND rail @57	33	black	GND rail link, B1 bottom to B2 top
☐	**W07**	B1 bottom 3V3 rail @59	B2 top 3V3 rail @59	41	red	3V3 rail link, B1 bottom to B2 top
☐	**W08**	B2 top GND rail @61	B2 bottom GND rail @61	51	black	GND rail link, B2 top to B2 bottom
	TFT (DS2)					
☐	**W09**	B1 b36	B1 top 3V3 rail @36	11	red	TFT V+ (Vin) 3V3
☐	**W10**	B1 b38	B1 top GND rail @37	13	black	TFT G (GND)
☐	**W11**	B1 b19	B1 c46	99	grey	GP21 to TFT BL (backlight PWM)
☐	**W12**	B1 b20	B1 d43	89	grey	GP20 to TFT RT (reset)
☐	**W13**	B1 b21	B1 e40	66	grey	GP19 SPI0 MOSI to TFT SI
☐	**W14**	B1 b22	B1 e39	53	grey	GP18 SPI0 SCK to TFT CK
☐	**W15**	B1 b24	B1 e42	58	grey	GP17 to TFT TC (chip select)
☐	**W16**	B1 b25	B1 e44	71	grey	GP16 to TFT DC
	Joystick (SW1)					
☐	**W17**	B1 b17	B1 b48	109	cyan	GP22 to joystick PRESS (pin 6)
☐	**W18**	B1 j16	B1 b49	130	cyan	GP8 to joystick UP (pin 5)
☐	**W19**	B1 b50	B1 top GND rail @49	13	black	Joystick COM (pin 4) to GND
☐	**W20**	B1 j24	B1 j48	64	cyan	GP14 to joystick LEFT (pin 1)
☐	**W21**	B1 j17	B1 j49	86	cyan	GP9 to joystick DOWN (pin 2)
☐	**W22**	B1 j25	B1 j50	66	cyan	GP15 to joystick RIGHT (pin 3)
	OLED (DS1)					
☐	**W23**	B1 j14	B1 e55	132	pink	GP6 I2C1 SDA to OLED Data
☐	**W24**	B1 j15	B1 e56	135	pink	GP7 I2C1 SCL to OLED Clk
☐	**W25**	B1 b61	B1 top 3V3 rail @60	11	red	OLED Vin 3V3
☐	**W26**	B1 b62	B1 top GND rail @61	13	black	OLED Gnd
☐	**W27**	B1 d57	B1 d62	15	black	OLED DC (= SA0) to GND, sets address 0x3C
	Pico signals to board 2					
☐	**W28**	B1 j9	B2 a6	57	yellow	GP2 bypass relay drive
☐	**W29**	B1 j7	B2 c28	117	green	GP1 UART0 RX from opto A (U1) output
☐	**W30**	B1 j6	B2 i30	152	blue	GP0 UART0 TX (OUT A)
☐	**W31**	B1 j11	B2 a47	145	violet	GP4 UART1 TX (OUT B)
	Relay drive					
☐	**W32**	B2 a9	B2 top GND rail @9	11	black	Q1 emitter to GND
☐	**W33**	B2 a11	B2 a13	8	yellow	COIL- : Q1 collector to D1 anode
☐	**W34**	B2 b13	B2 b16	10	yellow	COIL- : D1 anode to RL1 pin 16
☐	**W35**	B2 a16	B2 a40	69	yellow	COIL- : RL1 pin 16 to RL2 pin 1
☐	**W36**	B2 b40	B2 b42	8	yellow	COIL- : RL2 pin 1 to D2 anode
☐	**W37**	B2 j13	B2 bottom 5V rail @13	9	orange	D1 cathode to 5V
☐	**W38**	B2 j16	B2 bottom 5V rail @16	9	orange	RL1 coil + (pin 1) to 5V
☐	**W39**	B2 j40	B2 bottom 5V rail @40	9	orange	RL2 coil + (pin 16) to 5V
☐	**W40**	B2 j42	B2 bottom 5V rail @42	9	orange	D2 cathode to 5V
	MIDI IN A opto (U1)					
☐	**W41**	B2 c23	B2 g27	25	green	RL1 NO2 (pin 9) to H11L1 pin 2
☐	**W42**	B2 a26	B2 top 3V3 rail @27	9	red	H11L1 pin 6 Vcc to 3V3
☐	**W43**	B2 a27	B2 top GND rail @27	11	black	H11L1 pin 5 to GND
	Bypass links (relay NC to NC)					
☐	**W44**	B2 g21	B2 d35	53	brown	BYPASS 4: RL1 NC1 (pin 6) to RL2 NC1 (pin 6)
☐	**W45**	B2 a21	B2 g35	64	brown	BYPASS 5: RL1 NC2 (pin 11) to RL2 NC2 (pin 11)
	MIDI OUT B source					
☐	**W46**	B2 a51	B2 top 3V3 rail @51	6	red	3V3 to R8 (OUT B source)
	DIN flying leads (off-board)					
☐	**W47**	X1 pin 4	B2 j19	37	green	MIDI IN A DIN pin 4 to RL1 COM1 (pin 4)
☐	**W48**	X1 pin 5	B2 d19	71	green	MIDI IN A DIN pin 5 to RL1 COM2 (pin 13)
☐	**W49**	X2 pin 4	B2 d37	81	blue	OUT A DIN pin 4 to RL2 COM1 (pin 4)
☐	**W50**	X2 pin 5	B2 j37	39	blue	OUT A DIN pin 5 to RL2 COM2 (pin 13)
☐	**W51**	X2 pin 2	B2 bottom GND rail @39	30	black	OUT A DIN pin 2 (shield) to GND
☐	**W52**	X3 pin 4	B2 j51	36	violet	OUT B DIN pin 4 to R8 33R
☐	**W53**	X3 pin 5	B2 j47	36	violet	OUT B DIN pin 5 to R9 10R
☐	**W54**	X3 pin 2	B2 bottom GND rail @49	27	black	OUT B DIN pin 2 (shield) to GND (Rev B: was @53)
	MIDI IN B opto (U2, GP5) · Rev B					
☐	**W55**	B1 j12	B2 a57	168	white	GP5 UART1 RX from opto B (U2) output
☐	**W56**	B2 a55	B2 top 3V3 rail @55	6	red	H11L1 U2 pin 6 Vcc to 3V3
☐	**W57**	B2 a56	B2 top GND rail @55	11	black	H11L1 U2 pin 5 to GND
	MIDI IN B flying leads (off-board) · Rev B					
☐	**W58**	X4 pin 4	B2 j58	39	white	MIDI IN B DIN pin 4 to R10 220R
☐	**W59**	X4 pin 5	B2 j56	37	white	MIDI IN B DIN pin 5 to H11L1 U2 pin 2

Parts on the boards

Ref	Part / value	Pin → hole
A1	Raspberry Pi Pico 2	B1 straddles c/h, rows 6-25, USB end off the left edge. pin 1 GP0 = h6 ... pin 20 GP15 = h25; pin 21 GP16 = c25 ... pin 40 VBUS = c6.
R3	1k resistor GP2 to Q1 base	B2 b6 ⇔ b10
R4	10k resistor GP2 pull-down	B2 c6 ⇔ c9
Q1	2N3904	B2 E e9, B e10, C e11
D1	1N4148 diode RL1 flyback	B2 A d13, K g13
RL1	G5V-2-H1 5VDC	B2 1 f16, 4 f19, 6 f21, 8 f23, 16 e16, 13 e19, 11 e21, 9 e23
RL2	G5V-2-H1 5VDC	B2 8 e33, 6 e35, 4 e37, 1 e40, 9 f33, 11 f35, 13 f37, 16 f40
D2	1N4148 diode RL2 flyback	B2 A d42, K g42
U1	H11L1M	B2 1 f26, 2 f27, 3 f28, 6 e26, 5 e27, 4 e28
R1	220 Ω resistor opto A LED series	B2 i23 ⇔ i26
D3	1N4148 diode opto A LED reverse protection	B2 K j26, A j27
C1	100nF U1 decoupling	B2 b26 ⇔ b27
R2	470 Ω resistor opto A output pull-up	B2 b28 ⇔ top 3V3 rail @28
U2	H11L1M	B2 1 f55, 2 f56, 3 f57, 6 e55, 5 e56, 4 e57
R10	220 Ω resistor opto B LED series	B2 i58 ⇔ i55
D4	1N4148 diode opto B LED reverse protection	B2 K g55, A g56
C3	100nF U2 decoupling	B2 b55 ⇔ b56
R11	470 Ω resistor opto B output pull-up	B2 c57 ⇔ c55
R6	33 Ω resistor OUT A source	B2 b33 ⇔ top 3V3 rail @33
R7	10 Ω resistor OUT A sink	B2 j30 ⇔ j33
R9	10 Ω resistor OUT B sink	B2 d47 ⇔ g47
R8	33 Ω resistor OUT B source	B2 d51 ⇔ g51
DS1	Adafruit 938 1.3in OLED (I2C)	B1 header in column a: Data a55, Clk a56, DC a57, Rst a58, CS a59, 3v3 a60 Vin a61, Gnd a62
R5	10k resistor OLED RST pull-up	B1 c58 ⇔ c61
C2	100nF OLED RST to GND	B1 b58 ⇔ b57
DS2	Adafruit 5394 1.9in TFT	B1 header in column a: V+ a36, 3V a37, G a38, CK a39, SI a40, SO a41 TC a42, RT a43, DC a44, CC a45, BL a46
SW1	Adafruit 504 nav switch	B1 6(PRESS) d48, 5(UP) d49, 4(COM) d50 1(LEFT) f48, 2(DOWN) f49, 3(RIGHT) f50

Resistors and capacitors: either way round. Diodes: A = anode, K = cathode (band).
Relays: Omron G5V-2 terminal numbers (sheet 3). Q1: E, B, C.

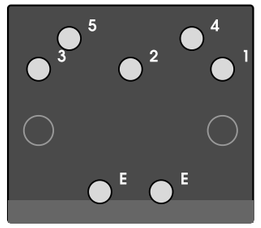
Sheet 3 · Pinouts for the parts that are easy to get wrong

5-pin DIN sockets X1-X4 · Same Sky SDS-50J (female, right-angle PCB)

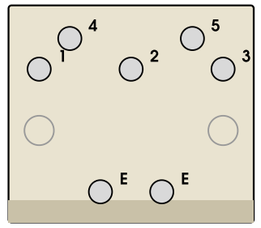
MIDI numbering runs 1-4-2-5-3 around the arc. Pin 2 (centre) = shield. Viewed from the front of a socket, pin 1 is on the RIGHT.
Pin positions taken from the Same Sky SDS-50J mechanical drawing (front view and recommended PCB layout).



FRONT (looking into the socket)
key notch at top; yellow 4/5 = loop, blue 2 = shield



↓ front / plug side ↓
TOP view (component side), as on sheet 1



↓ front / plug side ↓
SOLDER side (pins seen from below)

Lead per pin

X1 MIDI IN A

pin 4	B2 j19 (RL1 COM1)
pin 5	B2 d19 (RL1 COM2)
pin 2,1,3,E	NOT connected

X2 OUT A

pin 4	B2 d37 (RL2 COM1)
pin 5	B2 j37 (RL2 COM2)
pin 2	B2 GND rail @39
pin 1,3	NOT connected

X3 OUT B

pin 4	B2 j51 (R8 33Ω)
pin 5	B2 j47 (R9 10Ω)
pin 2	B2 GND rail @49
pin 1,3	NOT connected

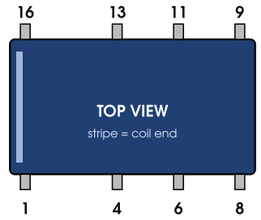
X4 MIDI IN B

pin 4	B2 j58 (R10 220Ω)
pin 5	B2 j56 (U2 pin 2)
pin 2,1,3,E	NOT connected

The shell tabs (E) are not used on the breadboard. Never ground a MIDI IN (A or B) shell or pin 2: the IN side must stay isolated (wiring.md section 2).
Socket pins are on a 2.5/5 mm grid, not 0.1": solder short (~15 cm) solid-core leads, sleeve each joint, label the socket.

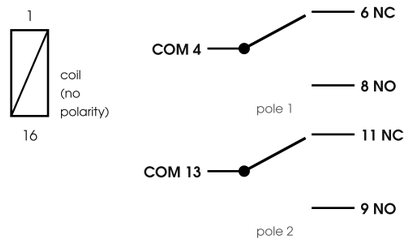
RL1, RL2 · Omron G5V-2-H1 5 VDC (DPDT)

From the Omron G5V-2 datasheet, 'Terminal arrangement / internal connections (bottom view)'.
VERIFY against your datasheet and with a meter before fitting.



Rows 0.3" apart: straddles the channel like a DIP-16.
Pin 1 bottom-left, 16 above it (DIP numbering).

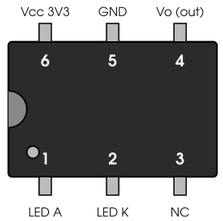
Internal (de-energised shown)



Pin	RL1 (IN side)
1/16	1 = +5V (j16), 16 = COIL-
4 / 13	COM: IN A DIN 4 / DIN 5
6 / 11	NC: to RL2 6 / RL2 11
8 / 9	NO: R1 220Ω / opto pin 2

RL2 (OUT A side, rotated 180°)
16 = +5V (j40), 1 = COIL-
COM: OUT A DIN 4 / DIN 5
NC: to RL1 6 / RL1 11
NO: R6 33Ω / R7 10Ω

U1 (IN A), U2 (IN B) · H11L1M, top view



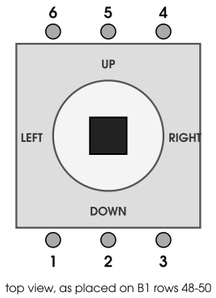
Notch (pin-1 dot) LEFT. B2 rows: U1 26-28, U2 55-57.
U1: 1 ← R1 220Ω ← RL1 NO1; 2 ← RL1 NO2; D3 K→1
U2: 1 ← R10 220Ω ← X4 pin 4; 2 ← X4 pin 5; D4 K→1
4 → GP1 (U1) / GP5 (U2) + 470Ω (R2 / R11) to 3V3
6 = 3V3 + 100nF (C1 / C3) to 5 = GND; 3 open

Q1 · 2N3904 (TO-92)



Legs spread to 0.1": E e9, B e10, C e11.
Flat face towards the channel.
E = GND, B ← 1k ← GP2, C = COIL-.

SW1 · Adafruit 504 5-way switch



top view, as placed on B1 rows 48-50

Pin	ALPS Function	GPIO strip		
6	Centre	PRESS	GP22	d48
5	D	UP	GP8	d49
4	Common	GND	-	d50
1	A	LEFT	GP14	f48
2	B	DOWN	GP9	f49
3	C	RIGHT	GP15	f50

Pin letters per the ALPS SKQUCAA010 data sheet that Adafruit links as 'similar'. Not on a 0.1" grid (3.25 mm): bend the legs to snap it in.
VERIFY each direction with a meter to pin 4.
Active low, internal pull-ups (no resistors).

Diodes and resistor colour codes

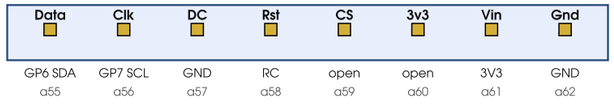


A (anode) K (cathode, band)

D1, D2 flyback: band (K) to +5V, i.e. B2 bottom half (g13, g42).
D3 opto A protection: band (K) in j26 (U1 pin 1), stand upright.
D4 opto B protection: band (K) in g55 (U2 pin 1), stand upright.

Value	Used for	4-band code (+ gold); 1% parts have 5 bands - measure
10 Ω	R7, R9	brown black black
33 Ω	R6, R8	orange orange black
220 Ω	R1, R10	red red brown
470 Ω	R2, R11	yellow violet brown
1 kΩ	R3	brown black red
10 kΩ	R4, R5	brown black orange
4.7 kΩ	(spare: I2C pull-ups)	yellow violet red

DS1 · Adafruit 938 1.3" OLED (I2C)



Before fitting, look at the back of the board:
• J1 and J2 must be CLOSED for I2C. Older 938s ship in SPI mode (bridge both with solder). The current STEMMA QT version ships with them closed already.
• Address: in I2C mode the DC pin is SA0. The firmware only probes 0x3C; Adafruit's 128x64 examples use 0x3D. W27 ties DC to GND to force 0x3C. Confirm with an I2C scan if it isn't detected.
• Rst: R5 10k to 3V3 + C2 100nF to GND (power-on reset). Harmless on the newer board, which resets itself.
• Power: Vin ← 3V3. The '3v3' pin is the on-board regulator OUTPUT: leave it open. CS: leave open in I2C mode.
• Module pull-ups on SDA/SCL are fitted; if yours has none add 4.7k from each to 3V3. SSD1306 is the firmware default.

DS2 · Adafruit 5394 1.9" ST7789 TFT



Label map (wiring.md → Adafruit silkscreen):
VCC → V+ (Vin), SCL/SCK → CK, SDA/MOSI → SI,
RES/RST → RT, DC/RS → DC, CS → TC, BLK/BL → BL.
SO (MISO) and CC (SD-card CS) are not used.
Header order above follows the Adafruit guide's pin list.
Check it against the silkscreen; if it differs, keep each wire on the strip under the matching label.
Detection: firmware probes the OLED at 0x3C first and uses the TFT only if nothing answers. Both may stay fitted; to run the TFT, pull W25 (OLED Vin) or build -DMIDIUM_DISPLAY=2.
Offset / colour order: TFT_X_OFFSET, TFT_Y_OFFSET, TFT_MADCTL in include/config.h.

A1 · Pico 2 pins used + Debug Probe

1	GP0	UART0 TX → OUT A	h6
2	GP1	UART0 RX ← opto A	h7
4	GP2	relay drive	h9
6	GP4	UART1 TX → OUT B	h11
7	GP5	UART1 RX ← opto B	h12
9/10	GP6/7	I2C1 SDA/SCL	h14/15
11/12	GP8/9	joy UP/DOWN	h16/17
19/20	GP14/15	joy LEFT/RIGHT	h24/25
21/22	GP16/17	TFT DC/CS	c25/24
24/25	GP18/19	SPI0 SCK/MOSI	c22/21
26/27	GP20/21	TFT RST/BL	c20/19
29	GP22	joy PRESS	c17
36	3V3(OUT)	3.3 V supply	c10
40	VBUS	5 V for relay coils	c6

Debug Probe 'D' port → SWD header at the far end of the Pico (opposite USB):
orange SC → SWCLK, black → GND,
yellow SD → SWDIO. Use the JST-to-JST cable (Pico 2 H) or JST-to-0.1" female (3-pin header).
The probe does not supply VBUS: keep the Pico USB plugged in or the relays never pull in.
Leave the 'U' port unconnected: UART0 is MIDI.
Reserved, not fitted: GP10-13 (microSD), GP26-28 (XRC/GPI), free: GP3.

Sheet 4 · Build order, bench test and later stages

Build order (milestone 1)

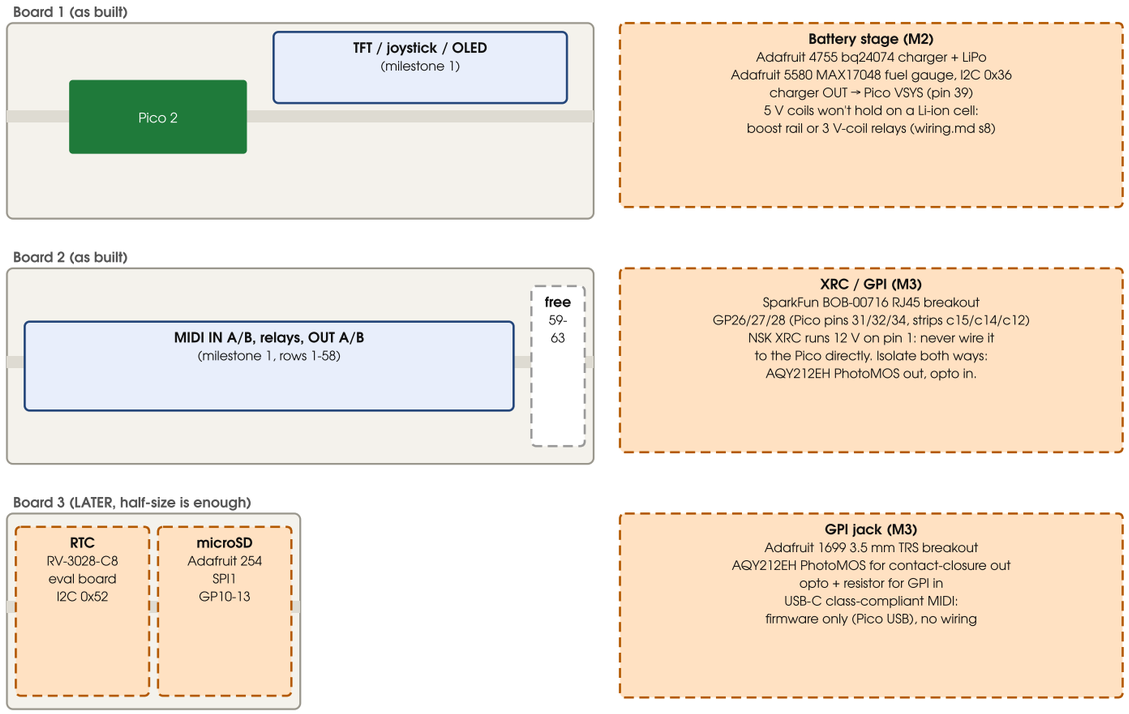
☐ **1. Boards**	Meter each rail end to end; bridge any split rail at the middle. Clip B1 above B2 as on sheet 1.
☐ **2. Pico + rails**	Fit A1 (USB left, c/h rows 6-25). W01-W04. USB in: B1 rails read 3.3 V and GND.
☐ **3. Supplies to B2**	W05 (VBUS), W06-W08. Check B2: top inner 3.3 V, bottom inner 5 V, outer rails GND.
☐ **4. Relay drive**	R3, R4, Q1, D1, D2, RL1, RL2 (stripes as drawn), W28, W32-W40. No coil current yet (GP2 low).
☐ **5. MIDI IN A**	U1 (notch left), R1, D3 upright, C1, R2, W29, W41-W43.
☐ **6. OUT A + bypass**	R6, R7, W30, W44, W45. Meter, unpowered: IN A DIN 4 ⇔ OUT A DIN 4 and 5 ⇔ 5 read ~0 Ω once leads are on.
☐ **7. OUT B**	R8, R9 across the channel, W31, W46.
☐ **8. MIDI IN B**	U2 (notch left, rows 55-57), R10, D4 upright (band in g55), C3, R11, W55-W57. No relay: IN B goes straight to U2.
☐ **9. DIN sockets**	Solder and sleeve the leads, then W47-W54 and W58-W59 (X4 IN B). X1/X4 pin 2 and shells stay free. Rev A builds: move W54 to the GND rail @49.
☐ **10. OLED**	Check J1/J2, then R5, C2, DS1, W23-W27.
☐ **11. TFT + joystick**	DS2, W09-W16; SW1 (meter-check directions), W17-W22.
☐ **12. Firmware**	Hold BOOTSEL, plug USB, copy midium.uf2 (or flash with the Debug Probe). Then the bench test below.

Bench test (docs/wiring.md section 9)

Needs: USB-MIDI interface (1 in / 1 out), a MIDI monitor + sender on the laptop, two DIN cables; a second interface or a synth on OUT A helps for step 3; step 8 needs a second MIDI source for IN B.

	#	Setup / action	Expected
☐	1	Unpowered. Laptop OUT → IN A, OUT A → 2nd interface/synth. Send notes.	Notes arrive at OUT A through the relays (passive bypass).
☐	2	Power from USB. Watch relays and screen.	Relays click ~1 s after power-up, splash, then 'MIDIum A→AB B→-'.
☐	3	Laptop OUT → IN A, OUT A → interface 2, OUT B → laptop IN. Send notes, CC, PC, bend, sysex.	Shown on screen, LED blinks, same data on A and B; sysex byte-identical on B.
☐	4	Same. Menu > Routing > Custom: IN A → OUT A off. Send notes.	Nothing on OUT A, OUT B still passes. Then Routing > Thru.
☐	5	Generate > Note test C4, CC, PC, MSC GO, MMC Play, PANIC.	Laptop sees each on OUT B; MSC GO = F0 7F 7F 02 7F 01 F7.
☐	6	Generate > Clock BPM 120 > Clock out on.	120.0 BPM measured (24 ppqn, 20.833 ms per clock).
☐	7	Dense stream 60 s on IN A: running-status CC sweep + 1 kB sysex; Note sweep + Clock out running.	Every byte on OUT B; Drops A/B = 0, UART errors = 0, Thru orphans = 0 (Mon overflow allowed).
☐	8	Second source into IN B. Menu > Routing > Merge (IN A + IN B → both). Send from both at once, incl. sysex on each.	Both streams merged on OUT A and OUT B, every message intact (sysex never interleaved). Monitor tags each line A or B.
☐	9	Pull USB power mid-stream.	Relays drop out; IN A hard-wired to OUT A again. IN B stops.

LATER · milestones 2 and 3 (not built yet)



- Where to tap the shared I2C1 bus (GP6 SDA / GP7 SCL): B1 holes i14 and i15 (Pico strips), or the OLED strips c55 / c56.
- Total bus pull-up: the OLED module already has some; add 4.7k to 3V3 only if a scope shows slow edges with RTC + gauge fitted.
- microSD (Adafruit 254): CLK ← GP10 (i19), DI ← GP11 (i20), DO → GP12 (i21), CS ← GP13 (i22); power 5V ← VBUS, GND.
- RTC eval board: 3V3, GND, SDA, SCL (+ INT/CLKOUT optional). pins.h says RV-3028-C7; the part ordered is the -C8 eval board.
- Fuel gauge: CELL/VIN to the LiPo, GND common, SDA/SCL to the bus. ALRT optional (spare GP3).
- Keep relay coil current in the battery budget: about 2 x 30 mA while running.

Verify before power-up (not settled by wiring.md)

- Relays: wiring.md names Omron G6K-2P-5VDC; the parts ordered are G5V-2-H1 5VDC. This layout uses the G5V-2 terminal numbers (coil 1-16, COM 4/13, NC 6/11, NO 8/9). Meter COM-NC = 0 ohm unpowered.
- OLED address: firmware probes 0x3C only; Adafruit 938 defaults to 0x3D unless DC (SA0) is low. W27 grounds DC. J1/J2 state depends on board revision.
- TFT header order and joystick pin letters are from Adafruit's guide / the ALPS data sheet it cites, not from the boards in hand.
- Pico 2 debug connector: 3-pin 0.1" header or JST-SH depending on the variant; pick the matching Debug Probe cable.

MIDIum MT-1	melling.audio
Breadboard prototype · Build order, bench test, later stages	Sheet 4 of 4
Rev C · 19/10/2026 · J Melling	A3 · milestone 1